Capstone release review

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete candidate gate and review-history record.

Keep manifest, conditions, pass/fail/not_run states, evidence references, blockers and remaining assurance limits together.

Keep one habit: Recall → W1–W3/W5 → independent access-failure case → release-review project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Capstone failure testing. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Documentation and operational readiness. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Performance and cost experiments. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- **Release review** combines correctness, access, failure recovery and operational readiness. **Critical gate**s can block release independently of an average score.

- Toy release requires all access tests, no unauthorised effects, at least 90% ordinary-case pass and a compatible **rollback rehearsal**. Record each gate separately with versioned **evidence**.

Capstone failure testing

> **My remembered idea, example and correction**
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Documentation and operational readiness

> **My remembered idea, example and correction**
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Performance and cost experiments

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Release review combines correctness, access, unauthorized-effect checks and compatible recovery under a declared candidate/version contract. Main r2 requires all access tests pass, no unauthorized effects, at least 90% ordinary pass rate and a compatible rollback rehearsal. Supplied 18/20 = 90% meets ordinary threshold; access/no-effects records pass, but rollback is not run, so not ready.

  Keep pass, fail, missing and not_run distinct. Separate picture p2 requires factual/access/deletion: first two pass, listed backup residual fails deletion, so blocked without majority voting. After a repair, current candidate evidence must support applicable gates; old p2 records cannot silently count for p3.

  New verification of all listed deletion locations and matching current pictured gate passes permits eligibility under that fictional set only. Preserve initial failure and each later decision. Constructed code evidence IDs/states are not real test executions or human approvals; no copies deleted or service deployed. Unknown stores remain outside listed-scope verification.

  Retain ownership, monitoring and recovery because approval cannot guarantee zero future faults.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

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  Keep pass, fail, missing and not_run distinct. Separate picture p2 requires factual/access/deletion: first two pass, listed backup residual fails deletion, so blocked without majority voting. After a repair, current candidate evidence must support applicable gates; old p2 records cannot silently count for p3.

  New verification of all listed deletion locations and matching current pictured gate passes permits eligibility under that fictional set only. Preserve initial failure and each later decision. Constructed code evidence IDs/states are not real test executions or human approvals; no copies deleted or service deployed. Unknown stores remain outside listed-scope verification.

  Retain ownership, monitoring and recovery because approval cannot guarantee zero future faults.

- Ordinary cases 18/20 pass; access all pass; no effects; rollback not tested. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Release review combines correctness, access, unauthorized-effect checks and compatible recovery under a declared candidate/version contract. Main r2 requires all access tests pass, no unauthorized effects, at least 90% ordinary pass rate and a compatible rollback rehearsal. Supplied 18/20 = 90% meets ordinary threshold; access/no-effects records pass, but rollback is not run, so not ready.

  Keep pass, fail, missing and not_run distinct. Separate picture p2 requires factual/access/deletion: first two pass, listed backup residual fails deletion, so blocked without majority voting. After a repair, current candidate evidence must support applicable gates; old p2 records cannot silently count for p3.

  New verification of all listed deletion locations and matching current pictured gate passes permits eligibility under that fictional set only. Preserve initial failure and each later decision. Constructed code evidence IDs/states are not real test executions or human approvals; no copies deleted or service deployed. Unknown stores remain outside listed-scope verification.

  Retain ownership, monitoring and recovery because approval cannot guarantee zero future faults.

- A report omits this qualification: Release readiness is scoped to the declared gates and environment; deployment still needs operational checks.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Release review combines correctness, access, unauthorized-effect checks and compatible recovery under a declared candidate/version contract. Main r2 requires all access tests pass, no unauthorized effects, at least 90% ordinary pass rate and a compatible rollback rehearsal. Supplied 18/20 = 90% meets ordinary threshold; access/no-effects records pass, but rollback is not run, so not ready.

  Keep pass, fail, missing and not_run distinct. Separate picture p2 requires factual/access/deletion: first two pass, listed backup residual fails deletion, so blocked without majority voting. After a repair, current candidate evidence must support applicable gates; old p2 records cannot silently count for p3.

  New verification of all listed deletion locations and matching current pictured gate passes permits eligibility under that fictional set only. Preserve initial failure and each later decision. Constructed code evidence IDs/states are not real test executions or human approvals; no copies deleted or service deployed. Unknown stores remain outside listed-scope verification.

  Retain ownership, monitoring and recovery because approval cannot guarantee zero future faults.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Ordinary 27/30 pass; access 1/5 fails; rollback passes; no effects. Decide.
- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- For Grounded Learning Service, use this week's independent task as a rehearsal for the final **evidence** pack: Ordinary 27/30 pass; access 1/5 fails; rollback passes; no effects. Decide.

  Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Capstone **release review**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Capstone **release review**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Ordinary 27/30 pass; access 1/5 fails; rollback passes; no effects. Decide.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Review every mandatory gate. A capstone release needs factual, access and deletion gates to pass.
- Factual and access checks pass. Deletion fails because a listed backup remains. All three gates are mandatory.
- Apply the three mandatory release gates to the recorded results. Why do two passes not outweigh the deletion failure, and what **evidence** must follow its repair?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Review every mandatory gate. A capstone release needs factual, access and deletion gates to pass.
- Which labelled step matches this detail? Remove the listed copy and verify it before a fresh review.
- 1. List **evidence**
- 2. Specify repair

- 3. Apply decision rule
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Specify repair Remove the listed copy and verify it before a fresh review. **Release review** applies explicit conditions to a specific candidate version.

Completed coding is not **evidence** that all operational obligations are satisfied.

Changed-case question and optional hint

- Changed-case practice: Every listed deletion location passes new verification. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. That gate can pass; review the complete current **evidence** before approving release.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: That gate can pass; review the complete current **evidence** before approving release. One repaired gate does not automatically refresh **evidence** for every other gate.

Repair and retain the record

Review Capstone failure testing (ai-6/34). Goal: Explain the **mechanism** and terminology of capstone failure testing. Short practice: Explain why retrieval success permits the checked 15:00 fallback when drafting fails.

What changes if retrieval fails too, and why can a template not fill that **evidence** gap?. Return to this week and explain what you changed.

Review Documentation and operational readiness (ai-6/32). Goal: Explain the **mechanism** and terminology of documentation and operational readiness. Short practice: Match SOURCE_UNAVAILABLE to its safe check and escalation owner.

What is lost when the maintainer route is removed, even if the diagnostic instructions stay readable?. Return to this week and explain what you changed.

Review Performance and cost experiments (ai-6/29). Goal: Explain the **mechanism** and terminology of performance and cost experiments. Short practice: Calculate per-case seconds and cost units for A and B.

Which meets a budget of at most 2 cost units per case, and what speed trade-off does that choice make?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Capstone release review: the case

AI ACADEMY / AI-6/35 · 01 · LOOK AT THE FACTS

Capstone release review: the case

The situation

Review every mandatory gate. A capstone release needs factual, access and deletion gates to pass.

Supplied evidence

Factual and access checks pass. Deletion fails because a listed backup remains. All three gates are mandatory.

Explain capstone release review using the deciding step in this fictional service.

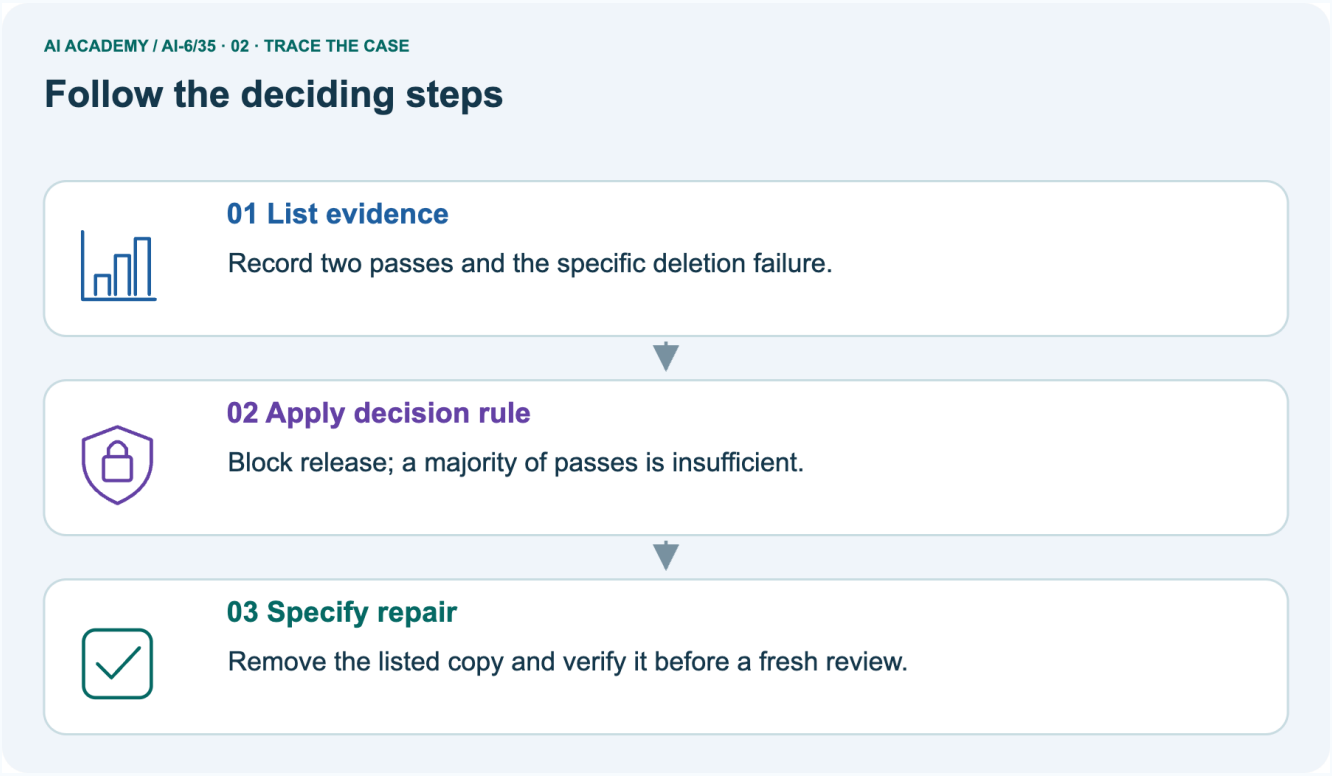
Follow the labelled picture with the matching explanation. Open the vector diagram

Review every mandatory gate. A capstone release needs factual, access and deletion gates to pass. Supplied facts: Factual and access checks pass. Deletion fails because a listed backup remains. All three gates are mandatory.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



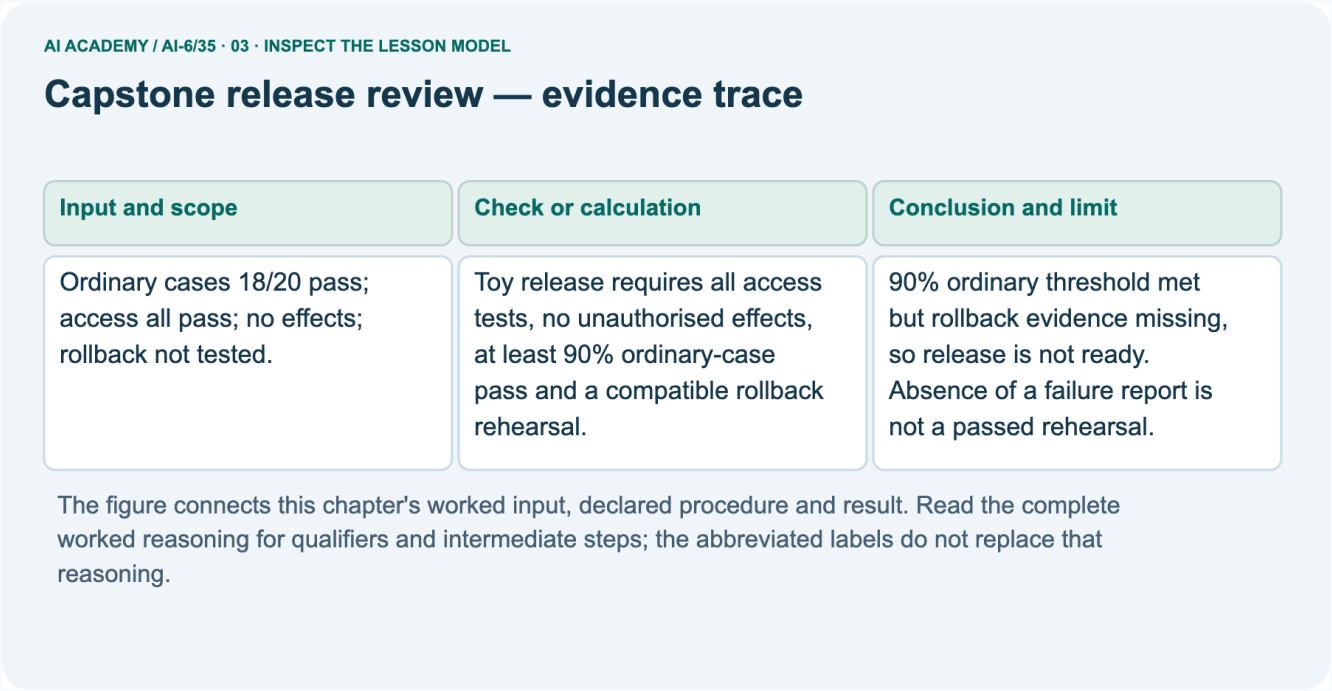
Follow the labelled picture with the matching explanation. Open the vector diagram

List **evidence**: Record two passes and the specific deletion failure. → Apply decision rule: Block release; a majority of passes is insufficient. → Specify repair: Remove the listed copy and verify it before a fresh review.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Capstone release review — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone **release review** — **evidence** trace.

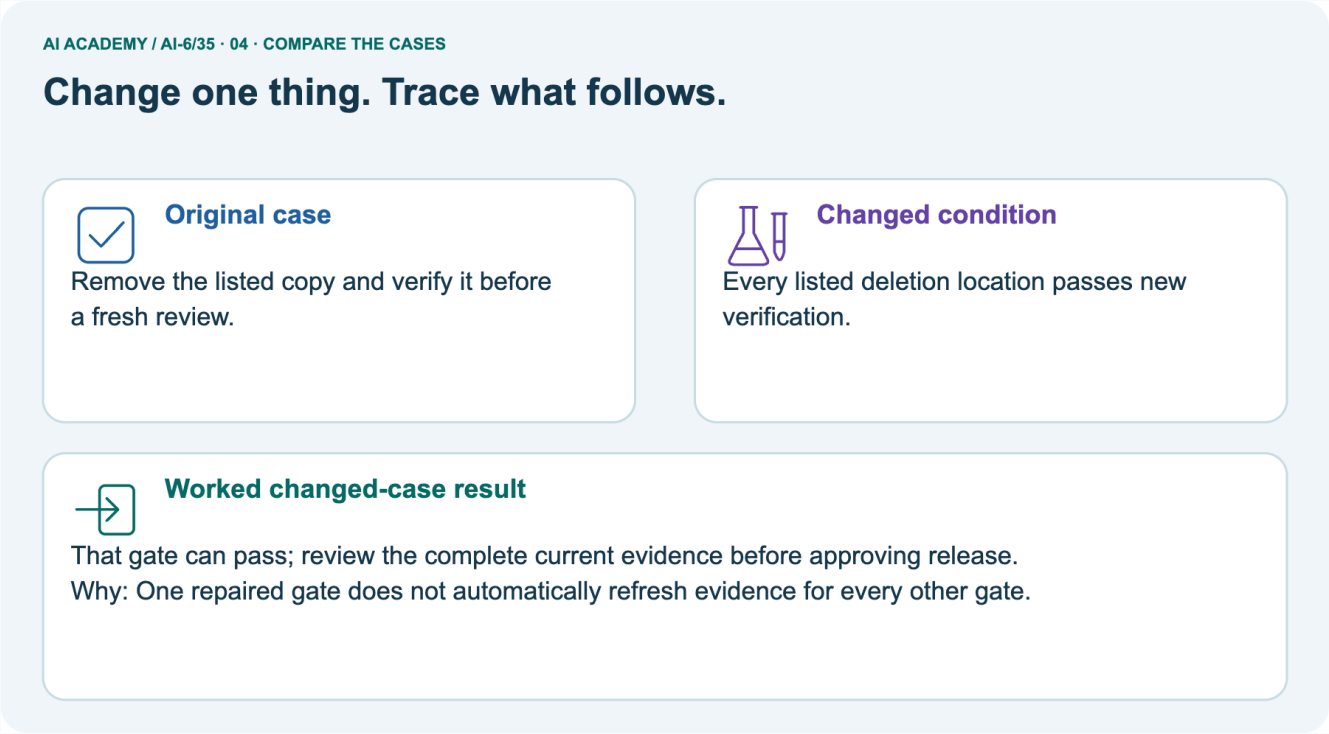
Ordinary cases 18/20 pass; access all pass; no effects; rollback not tested. | Toy release requires all access tests, no unauthorised effects, at least 90% ordinary-case pass and a compatible **rollback rehearsal**. | 90% ordinary threshold met but rollback **evidence** missing, so release is not ready.

Absence of a failure report is not a passed rehearsal.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

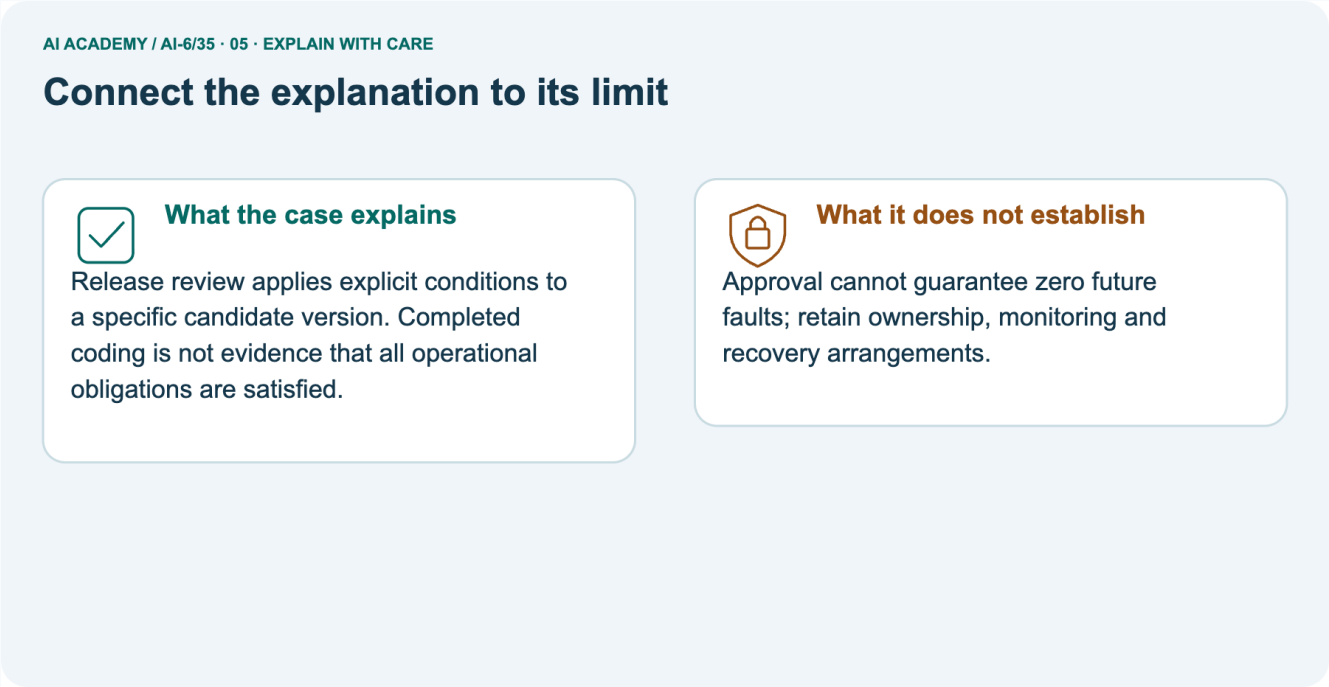
Original case: Remove the listed copy and verify it before a fresh review. Changed condition: Every listed deletion location passes new verification.

Expected result: That gate can pass; review the complete current **evidence** before approving release. Reason: One repaired gate does not automatically refresh **evidence** for every other gate.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

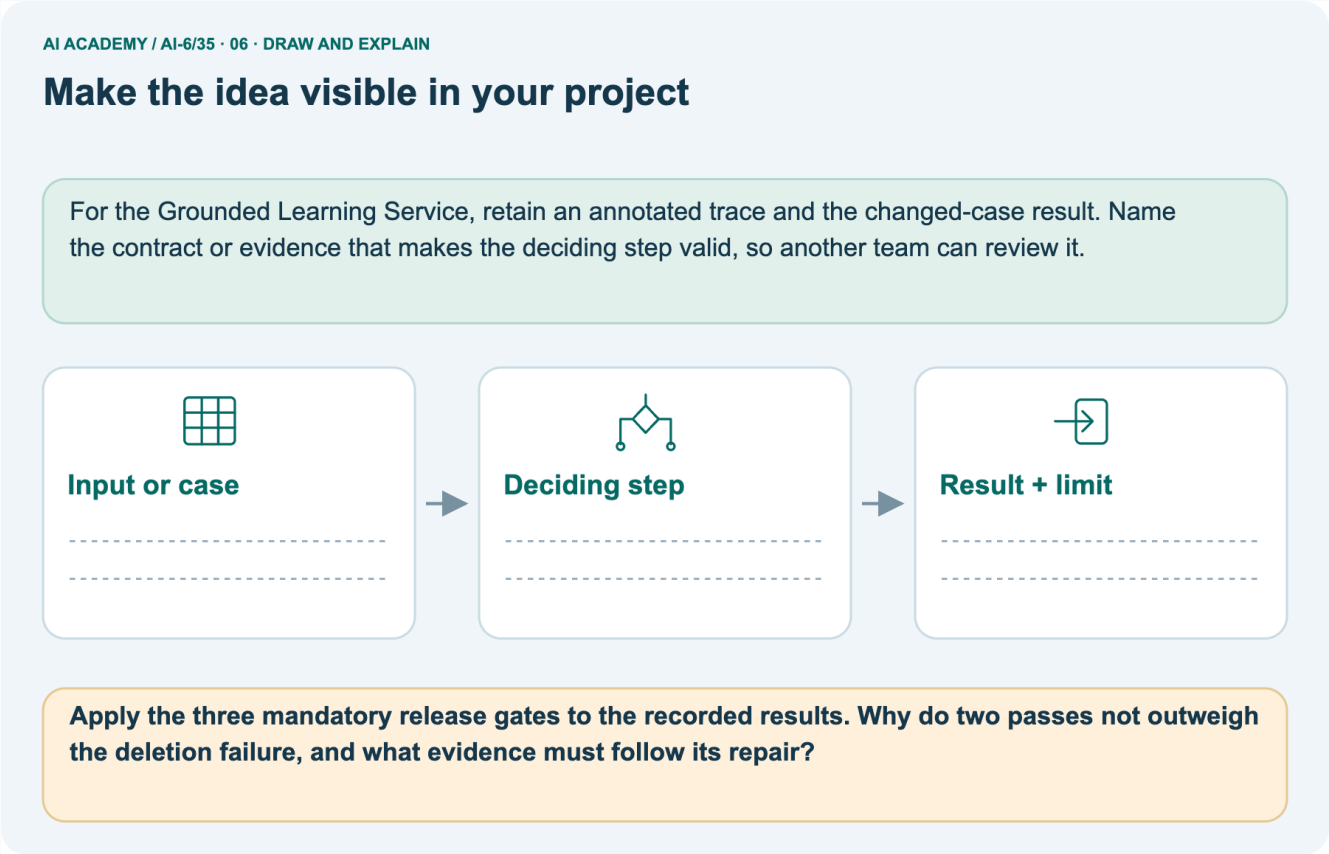
Release review applies explicit conditions to a specific candidate version. Completed coding is not **evidence** that all operational obligations are satisfied.

Limit: Approval cannot guarantee zero future faults; retain ownership, monitoring and recovery arrangements.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Apply the three mandatory release gates to the recorded results.

Why do two passes not outweigh the deletion failure, and what **evidence** must follow its repair?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Capstone release review: the case

02. Follow the deciding steps

03. Capstone release review — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.